Negative Exponents & Scientific Notation

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Rules to remember

$a^{-n} = \frac{1}{a^n}$

$\frac{1}{a^{-n}} = a^n$

$a^m \cdot a^n = a^{m+n}$

$\frac{a^m}{a^n} = a^{m-n}$

$(a^m)^n = a^{mn}$

Scientific notation: $a \times 10^n$

where $1 \le a < 10$

Example: scientific notation

To scientific: $0.00052 = 5.2 \times 10^{-4}$

(decimal moved 4 places right, so exponent is −4)

From scientific: $3.1 \times 10^5 = 310{,}000$

(exponent is 5, so move decimal 5 places right)

Number ≥ 10: positive exponent. Number < 1: negative.

Rewrite using only positive exponents.

1. 9^{-3}

Answer: _____

2. $5x^{-2}y^3$

Answer: _____

Rewrite using a negative exponent.

3. $\frac{1}{8^5}$

Answer: _____

4. $\frac{3}{x^4}$

Answer: _____

Evaluate. Write your answer as a whole number or a fraction.

5. 4^{-2}

Answer: _____

6. $(-2)^{-3}$

Answer: _____

Simplify. Write your answer using only positive exponents.

7. $y^{-4} \cdot y^{9}$

Answer: _____

8. $3^{4} \cdot 3^{-7}$

Answer: _____

9. $\dfrac{a^{-5}}{a^{2}}$

Answer: _____

10. $\dfrac{x^{-1}}{x^{-6}}$

Answer: _____

11. $(y^{4})^{-2}$

Answer: _____

12. $(2^{-3})^{-1}$

Answer: _____

Write in scientific notation.

13. 45,000

Answer: _____

14. 6,030,000

Answer: _____

15. 0.0072

Answer: _____

16. 0.000108

Answer: _____

Write in standard form.

17. 3.2×10^{5}

Answer: _____

18. 7.01×10^{3}

Answer: _____

19. 4×10^{-3}

Answer: _____

20. 5.6×10^{-5}

Answer: _____